Assignment No	7
Title	**Circular Queue**
Objective	learning and understanding
Roll No	MCA2649

Source Code:-

```
class CircularQueue
{
    int[] data;
    int front;
    int rear;
    int capacity;

    //define constructor
    CircularQueue(int size)
    {
        this.capacity=size;
        data = new int[capacity];
        rear=-1;
        front=-1;
    }

    //insert an element
```

```
void enqueue(int ele)
{
        //check whether queue is full
        if((rear+1)% capacity == front)
        {
                System.out.println("Queue is Full");
                return;
        }
        //insert first element
        if(front==-1)
        {
                front=0;
                rear=0;
        }
        // any other element including wrap around
        else
        {
                rear=(rear+1)% capacity;
        }
        // store the value
        data[rear]=ele;
        System.out.println("Element Added:"+ele);
}

//remove an element
```

```
void dequeue()
{
        // check whethe the queue us empty
        if(front== -1)
        {
                System.out.println("Queue is Full");
                return;
        }
        // remove the element
        System.out.println("Element removed:"+data[front]);
        // update the value of front
        if(front==rear)
        {
                front=-1;
                rear=-1;
        }
        else
        {
                front=(front+1)%capacity;
        }
}
//print
void display()
{
        if(front== -1)
```

```
            {
                System.out.println("Queue is Full");
                return;
            }
            System.out.println("Queue elements....");
            int i=front;
            while(true)
            {
                System.out.print(data[i]+" ");
                if(i==rear)
                {
                        break;
                }
                i=(i+1)         %capacity;
            }
            System.out.println();
        }

}
class  CircularQueueImpl
{
        public static void main(String args[])
        {
            //create an object of queue
            CircularQueue c1 = new CircularQueue(4);
```

```
            for(int i=0;i<=4;i++)
            {
                c1.enqueue(i*10);
                c1.display();
            }
            System.out.println("***** Dequeue *****");
            c1.dequeue();
            c1.dequeue();
            c1.display();

            System.out.println("***** Circular Operation *****");
            c1.enqueue(40);
            c1.display();

            c1.enqueue(50);
            c1.display();

            c1.enqueue(60);
        }

}
```

OUTPUT:-

```
D:\MCA2649\ADS2649>java CircularQueueImpl
Element Added:0
Queue elements....
0
Element Added:10
Queue elements....
0 10
Element Added:20
Queue elements....
0 10 20
Element Added:30
Queue elements....
0 10 20 30
Queue is Full
Queue elements....
0 10 20 30
***** Dequeue *****
Element removed:0
Element removed:10
Queue elements....
20 30
***** Circular Operation *****
Element Added:40
Queue elements....
20 30 40
Element Added:50
Queue elements....
20 30 40 50
Queue is Full

D:\MCA2649\ADS2649>
```